

Unit - 5. P.D.E. - Higher Order

Classification of P.D.E. of second order $\Rightarrow$

$$\text{Gen. form} \Rightarrow A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial t} + C \frac{\partial^2 u}{\partial t^2} + F\left(x, t, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial t}\right) = 0$$

where A, B, C are function of x & t .

Now eq. (1) can be classified in three categories:

(i) Elliptic if $B^2 - 4AC < 0$

(ii) Parabolic if $B^2 - 4AC = 0$

(iii) Hyperbolic if $B^2 - 4AC > 0$

Example $\Rightarrow$

① one dimensional wave eq. $\Rightarrow$

$$c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad (\text{RTU-2024})$$

① is hyperbolic as $B^2 - 4AC = 0 - 4(c^2)(-1) = 4c^2 > 0$

② one dimensional Heat eq. $\Rightarrow$

$$c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} \quad (\text{RTU-2023, 2021})$$

is parabolic as $B^2 - 4AC = 0 - 4(c^2)(0) = 0$

③ Two dimensional Laplace's eq. $\Rightarrow$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

is elliptic as $B^2 - 4AC = 0 - 4(1)(1) = -4 < 0$

Method of separation of variable $\Rightarrow$ (RTU-2023)

let u be a fn of two independent variables x & t .

$$u(x, t) = X(x) T(t) \quad \text{--- (1)}$$

$$X(x) \Rightarrow \text{fn of } x \text{ alone} \Rightarrow \frac{dX}{dx} = 0$$

$$T(t) \Rightarrow \text{fn of } t \text{ alone} \Rightarrow \frac{dT}{dt} = 0$$

$$\text{Now, } \frac{du}{dx} = T X' \quad \text{and} \quad \frac{du}{dt} = X T' \quad \text{--- (2)}$$

Substitute the values of $\frac{du}{dx}$ and $\frac{du}{dt}$ in given p.d.e,
which produces

$$f(x, x', x'', \dots) = g(t, t', t'', \dots) = k \quad (\text{let})$$

$$\text{now, } f(x, x', x'', \dots) = k \quad \text{--- (3)}$$

$$\text{and } g(t, t', t'', \dots) = k \quad \text{--- (4)}$$

now eq. (3) and eq. (4) are o.d.e. can be solved by known methods. we get X and T .

now substitute values of X and T in eq. (1).

we get desired solution.

Q: solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ by method of separation of variables where $u(x, 0) = 6e^{3x}$. (RTU-2024) (5-m)

Sol: Given that $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ — (1)

Let the sol. of eq. (1) $\Rightarrow u(x, t) = X(x) \cdot T(t) \Rightarrow u = XT$

now $\frac{\partial u}{\partial x} = T X'$ and $\frac{\partial u}{\partial t} = X T'$ — (2)

Put the value of eq. (2) in eq. (1),

$$X' T = 2 X T' + X T$$

$$\Rightarrow X' T - X T = 2 X T'$$

$$\Rightarrow T (X' - X) = 2 X T'$$

$$\Rightarrow \frac{X' - X}{X} = \frac{2 T'}{T} = k$$

now, $\frac{X' - X}{X} = k$

and $\frac{2 T'}{T} = \frac{k}{2}$

$$\frac{X'}{X} - 1 = k$$

$$\Rightarrow \log T = \frac{k}{2} t + \log c_1$$

$$\log X - X = k X$$

$$\Rightarrow T = c_1 e^{\frac{k}{2} t}$$

$$\log X = X(k+1)$$

$$X = e^{X(k+1)}$$

So, $u(x,t) = c_1 e^{(k+1)x} \cdot c_2 e^{kt/2}$

$$u(x,t) = c e^{(k+1)x + \frac{kt}{2}} \quad \text{--- (3)}$$

Put $t=0$

$$u(x,0) = 6 e^{-3x} = c e^{(k+1)x}$$

$$\Rightarrow c=6 \quad \text{and} \quad k+1=-3$$

$$k=-4$$

So, from eq. (3)

$$u(x,t) = 6 e^{-3x-2t}$$

$$\boxed{u(x,t) = 6 e^{-(3x+2t)}}$$

Ans .

Solution of Laplace Equations in two dimension $\Rightarrow$ (RTU-2024, 23, 2021) (10-M)

The sol. of Laplace eq, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ — (1)

Can be obtained by method of separation of variable,

let sol of eq. (1) $\Rightarrow u(x, y) = x(x) y(y)$

$$\begin{aligned} \text{then } \frac{\partial^2 u}{\partial x^2} &= y \frac{\partial^2 x}{\partial x^2} \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = x \frac{\partial^2 y}{\partial y^2} \\ &= y \frac{d^2 x}{dx^2} \quad \quad \quad = x \frac{d^2 y}{dy^2} \end{aligned}$$

so, from eq. (1)

$$y \frac{d^2 x}{dx^2} + x \frac{d^2 y}{dy^2}$$

$$\Rightarrow y \frac{d^2 x}{dx^2} = -x \frac{d^2 y}{dy^2}$$

$$\Rightarrow \frac{1}{x} \frac{d^2 x}{dx^2} = -\frac{1}{y} \frac{d^2 y}{dy^2} = k \quad \text{--- (2)}$$

$\uparrow$ Constant

Then $\frac{d^2 x}{dx^2} - kx = 0$ — (3)

and $\frac{d^2 y}{dy^2} + ky = 0$ — (4)

If $k \geq 0 \Rightarrow$ sol. of eq. (3) $\Rightarrow x = A e^{\sqrt{k}x} + B e^{-\sqrt{k}x}$

$$x = A \cosh \sqrt{k}x + B \sinh \sqrt{k}x \quad (\text{or})$$

and sol. of eq. (4)

$$y = A \cos \sqrt{k}y + B \sin \sqrt{k}y$$

$\therefore$ solution of eq. (1) is $\Rightarrow u = xy$

$$u = (A e^{\sqrt{k}x} + B e^{-\sqrt{k}x}) (A \cos \sqrt{k}y + B \sin \sqrt{k}y)$$

$$\text{or } u = (A \cosh \sqrt{k}x + B \sinh \sqrt{k}x) (A \cos \sqrt{k}y + B \sin \sqrt{k}y) \quad \text{--- (5)}$$

If $k < 0$ (let $k = -\lambda^2$)

$$\text{eq. (3)} \Rightarrow \frac{d^2x}{dy^2} + \lambda^2 x = 0$$

$$\text{sol.} \Rightarrow x = A \cos \lambda x + B \sin \lambda x$$

$$\text{eq. (4)} \Rightarrow \frac{d^2y}{dx^2} - \lambda^2 y = 0$$

$$\text{sol.} \Rightarrow y = C e^{\lambda y} + D e^{-\lambda y}$$

$\therefore$ solution of eq. (1) is $\Rightarrow u = xy$

$$u = (A \cos \lambda x + B \sin \lambda x) (C e^{\lambda y} + D e^{-\lambda y}) \quad \text{--- (6)}$$

The nature of solution will depend on the boundary conditions given in the problem.

One Dimensional Wave Eq. $\Rightarrow$

Q: A string is stretched between the fixed points $(0,0)$ and $(l,0)$ and released from rest from position $y = A \sin \frac{\pi x}{l}$.

Find $y(x,t)$.

Sol: The d.e of wave eq. $\Rightarrow \frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{--- (1)}$

let the sol. of (1) $\Rightarrow y = x(x) T(t)$.

$y = XT \quad \text{--- (2)}$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = x \frac{d^2 T}{dt^2} \quad \text{and} \quad \frac{\partial^2 y}{\partial x^2} = T \frac{d^2 x}{dx^2}$$

$\therefore$ Eq. (1) gives

$$x \frac{d^2 T}{dt^2} = c^2 T \frac{d^2 x}{dx^2}$$

$$\Rightarrow \frac{1}{c^2 T} \frac{d^2 T}{dt^2} = \frac{1}{x} \frac{d^2 x}{dx^2} = -k^2$$

then $\frac{1}{c^2 T} \frac{d^2 T}{dt^2} = -k^2 \quad \text{and} \quad \frac{1}{x} \frac{d^2 x}{dx^2} = -k^2$

$$\Rightarrow \frac{d^2 T}{dt^2} + c^2 k^2 T = 0$$

$$\frac{d^2 x}{dx^2} + k^2 x = 0$$

whose solutions are

$$T = A_1 \cos kct + A_2 \sin kct$$

and $x = A_3 \cos kx + A_4 \sin kx$

eq. (2) gives $\Rightarrow y = (A_1 \cos kct + A_2 \sin kct) (A_3 \cos kx + A_4 \sin kx) \quad \text{--- (3)}$

Given boundary condition $\Rightarrow$

$$x=0, y=0 \quad \text{and} \quad x=l, y=0$$

└─ (a)

└─ (b)

Using (a) in eq. (3),

$$0 = (A_1 \cos ckt + A_2 \sin ckt) A_3 \Rightarrow A_3 = 0 \text{ — (4)}$$

Using (b) and (4) in eq. (3)

$$0 = (A_1 \cos ckt + A_2 \sin ckt) A_4 \sin kl$$

$$\Rightarrow \sin kl = 0$$

($\because A_4 \neq 0$ as $A_3 = 0 = A_4$ gives $x=0$ which is not possible),

$$\Rightarrow kl = n\pi \Rightarrow k = \frac{n\pi}{l}, \quad n \in \mathbb{I}$$

$$\therefore \text{eq. (3) becomes } \Rightarrow y = \left(A_n \cos \frac{n\pi c}{l} t + B_n \sin \frac{n\pi c}{l} t \right) \sin \frac{n\pi x}{l} \text{ — (5)}$$

where $A_n = A_1 A_4$ and $B_n = A_2 A_4$.

~~As~~ As different solutions are possible for varying values of n , the most general solution is obtained by adding all these solutions.

$$y(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos \frac{n\pi c}{l} t + B_n \sin \frac{n\pi c}{l} t \right] \sin \frac{n\pi x}{l} \text{ — (6)}$$

Applying initial condition

Cond. (1) $\Rightarrow y = A \sin \frac{\pi x}{l}$ for $t=0$ in eq. (6)

we get

$$A \sin \frac{\pi x}{l} = \sum_{n=0}^{\infty} A_n \sin \frac{n\pi x}{l}$$

$$A \sin \frac{\pi x}{l} = A_1 \sin \frac{\pi x}{l} + A_2 \sin \frac{2\pi x}{l} + \dots$$

on Comparing

$$A = A_1 \quad \text{and} \quad A_2 = A_3 = \dots = 0 \quad \text{--- (7)}$$

Cond. (2) $\frac{\partial y}{\partial t} = 0$ for $t = 0$

eq. (6) become $\Rightarrow \frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} \left[-\frac{n\pi c}{l} A_n \sin \frac{n\pi x}{l} + \frac{n\pi c}{l} B_n \cos \frac{n\pi x}{l} \right] \sin \frac{n\pi t}{l}$

using Cond (2) $\Rightarrow 0 = \sum_{n=1}^{\infty} \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}$

$$\Rightarrow B_1 = B_2 = \dots = 0 \quad \Rightarrow B_n = 0 \quad \text{--- (8)}$$

using eq. (7) & (8) in eq. (6),

we get $y(x, t) = A \cos \frac{\pi c t}{l} \sin \frac{\pi x}{l}$ Ans.

One Dimensional Heat Eq. $\Rightarrow$

$$\text{Heat eq.} \Rightarrow \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{--- (1)}$$

$$\text{let solution of eq. (1)} \Rightarrow u(x, t) = X(x) T(t) \quad \text{--- (2)}$$
$$u = XT$$

$$\Rightarrow \frac{\partial u}{\partial t} = X \frac{dT}{dt} \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} = T \frac{d^2 X}{dx^2}$$

$\therefore$ Eq. (1) reduces $\Rightarrow$

$$X \frac{dT}{dt} = c^2 T \frac{d^2 X}{dx^2}$$

$$\Rightarrow \frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{c^2 T} \frac{dT}{dt} = K$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} = K$$

$$\frac{d^2 X}{dx^2} - KX = 0$$

$$\frac{1}{c^2 T} \frac{dT}{dt} = K$$

$$\frac{dT}{dt} - Kc^2 T = 0$$

Case-1 $\boxed{K=0}$

$$\frac{d^2 X}{dx^2} = 0$$

$$\Rightarrow X = Ax + B$$

$$\frac{dT}{dt} = 0$$

$$T = C$$

$$\Rightarrow u(x, t) = (Ax + B)(C)$$

Case-2

$$K = \lambda^2 > 0$$

$$\frac{d^2 x}{dx^2} - \lambda^2 x = 0$$

$$x = A e^{\lambda x} + B e^{-\lambda x}$$

$$\frac{1}{T} \frac{dT}{dt} = \lambda^2 c^2$$

$$\log_e T = \lambda^2 c^2 t + \log D$$

$$T = D e^{\lambda^2 c^2 t}$$

$$u(x, t) = (A e^{\lambda x} + B e^{-\lambda x}) (D e^{\lambda^2 c^2 t})$$

Case-3

$$K = -\lambda^2 < 0$$

$$\frac{d^2 x}{dx^2} + \lambda^2 x = 0$$

$$x = A \cos \lambda x + B \sin \lambda x$$

$$\frac{dT}{dt} = -\lambda^2 c^2 T$$

$$T = D e^{-\lambda^2 c^2 t} + \log D$$

$$T = D e^{-\lambda^2 c^2 t}$$

$$u(x, t) = (A \cos \lambda x + B \sin \lambda x) (D e^{-\lambda^2 c^2 t})$$